

Seunghwan Lee

Human-Centered Interactive Systems
with AI, Data, Sensing, and Prototyping

Introduction

My approach to technology is clear: technology created by people should ultimately serve people.

I have focused on how people experience limitations in accessing information, interacting with technology, and regulating everyday behavior, and on building interactive systems that respond to those situations. My choice to study across Systems Biotechnology, Photography and New Media, and Nano-Convergence Materials Engineering came from an interest that gradually expanded from understanding living systems, to understanding human perception and lived experience, and finally to exploring materials and physical interaction. These perspectives eventually converged in my interest in building human-centered interactive systems.

Across my projects, I have built systems that connect human behavior and sensory experience with computation, including AI-assisted accessibility tools, tangible interactions that transform physical behavior into data, digital wellbeing systems, and data-driven interactive visualizations. I have also developed registered interaction concepts involving wearable sensing, passive physical input, and gesture-based interaction.

Through these experiences, I became increasingly interested in human-centered AI and AI × HCI: how AI and computational systems can interpret human behavior, context, and needs, and how those capabilities can be translated into interactions that people can meaningfully use in everyday life. This interest is grounded in a broader concern for people whose needs, abilities, and lived experiences are not always well reflected in conventional technology design. I am particularly interested in accessibility, digital wellbeing, multimodal interaction, and data-driven interactive systems.

My background in physical computing and prototyping remains an important part of this direction. Rather than treating physical interfaces as an end in themselves, I see sensing, tangible interaction, and multimodal feedback as ways of connecting intelligent systems to human behavior and lived experience.

All projects in this portfolio were individually conceived and implemented unless otherwise noted.

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Part I.

Human-Centered AI & Interactive Systems

: Working systems that connect AI, sensing, data, and human behavior

I Can See 2

RQ. How can visual images become accessible through non-visual sensory experiences?



Prototype v1

Initial structure for converting AI-generated image descriptions into Korean Braille and printing them through the BL-1000 Braille Printer.

01 Motivation & Problem

- I focused on the limited access to visual media that blind and low-vision users experience in a visual-centered environment.
- Conventional printed Braille materials deliver fixed information, making it difficult to respond flexibly to newly introduced image-based content.
- To address this, I proposed a multisensory assistive system that translates image analysis results into Braille and audio access.

02 System Architecture

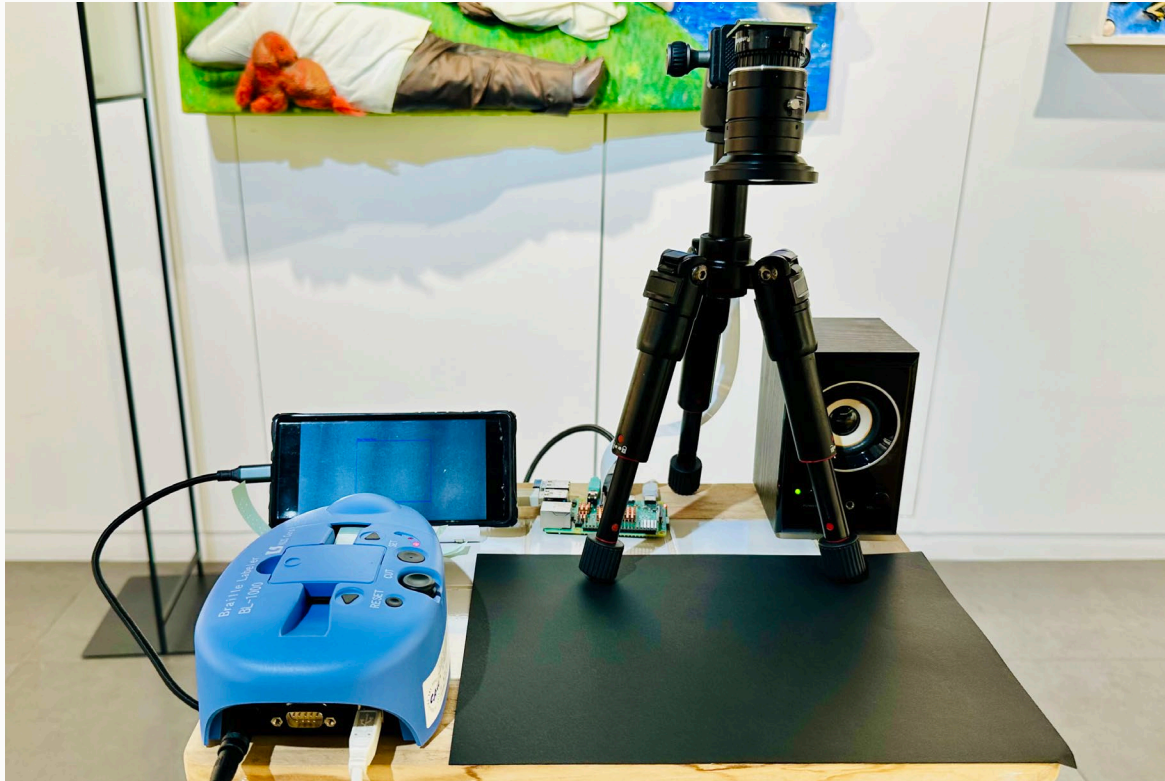
- **Image Input & AI Parsing:** The system captures or receives visual images and analyzes scene and object information through AI.
- **Output Separation:** The analysis result is separated into two output layers: a detailed audio description and a concise Braille-oriented summary.
- **Braille & Speech Output:** The Braille-oriented text is converted based on the 2024 revised Korean Braille regulations and physically printed, while the audio output is provided through TTS.

03 Cooperation

The initial hardware prototype was refined through formative accessibility feedback from staff at Chung-Ang University's Disability Student Support Center.

I Can See 2

RQ. How can visual images become accessible through non-visual sensory experiences?



Prototype v2

Expanded prototype that separates image analysis results into audio descriptions and Braille summaries for auditory and tactile access.

04 Design Decision

- Rather than simply converting images into text, this project treats language as an intermediate layer for translating visual media into tactile and auditory forms.
- Audio output provides richer scene description, while Braille output uses shorter summaries suited to tactile reading.
- For Braille conversion, I implemented a custom Korean Braille algorithm based on the 2024 Revised Korean Braille Regulations, instead of relying on an external translator.

05 Evaluation & Next Step

- The prototype was demonstrated to staff at the Siloam Center for the Blind for preliminary accessibility feedback.
- The feedback suggested that Braille output should remain concise due to the physical length limits of printed Braille, while audio descriptions should provide richer spatial information, such as where each object is located.
- I also received the 2025 Tactile Graphic Material Production Guidelines from the National Library for the Disabled as a reference for further development, and plan to refine the system based on these guidelines.

06 Contribution

This was an individual project covering problem definition, system planning, hardware setup, AI integration, Korean Braille conversion, Braille printing, TTS output, prototyping, and documentation.

Visual-to-Tactile & Audio Translation Pipeline



HCI Design Points

- Visual media as transformable sensory information
- Language as an intermediate layer between visual, auditory, and tactile senses
- Different information density for audio and Braille outputs
- Physical Braille printing as tangible accessible feedback
- User-provided phone images made accessible through multisensory output

Implementation

- Raspberry Pi 4B + Pi HQ Camera
- OpenCV phone-screen detection and image-region extraction
- OpenAI-based image interpretation and TTS
- Python-based Korean Braille conversion to Braille ASCII
- BL-1000 Braille Printer output



Working Prototype v1: initial hardware setup and Braille output generated through AI image interpretation and Korean Braille conversion.



► DEMO VIDEO

https://youtu.be/UA_z8u3LI_Q

Demo video of the initial workflow: Visual input → AI interpretation → Braille printing.



Prototype v2 incorporating higher-resolution image capture and audio output after formative accessibility feedback.



Formative demonstration with the Tactile Teaching Materials Production Team at the Siloam Center for the Blind.



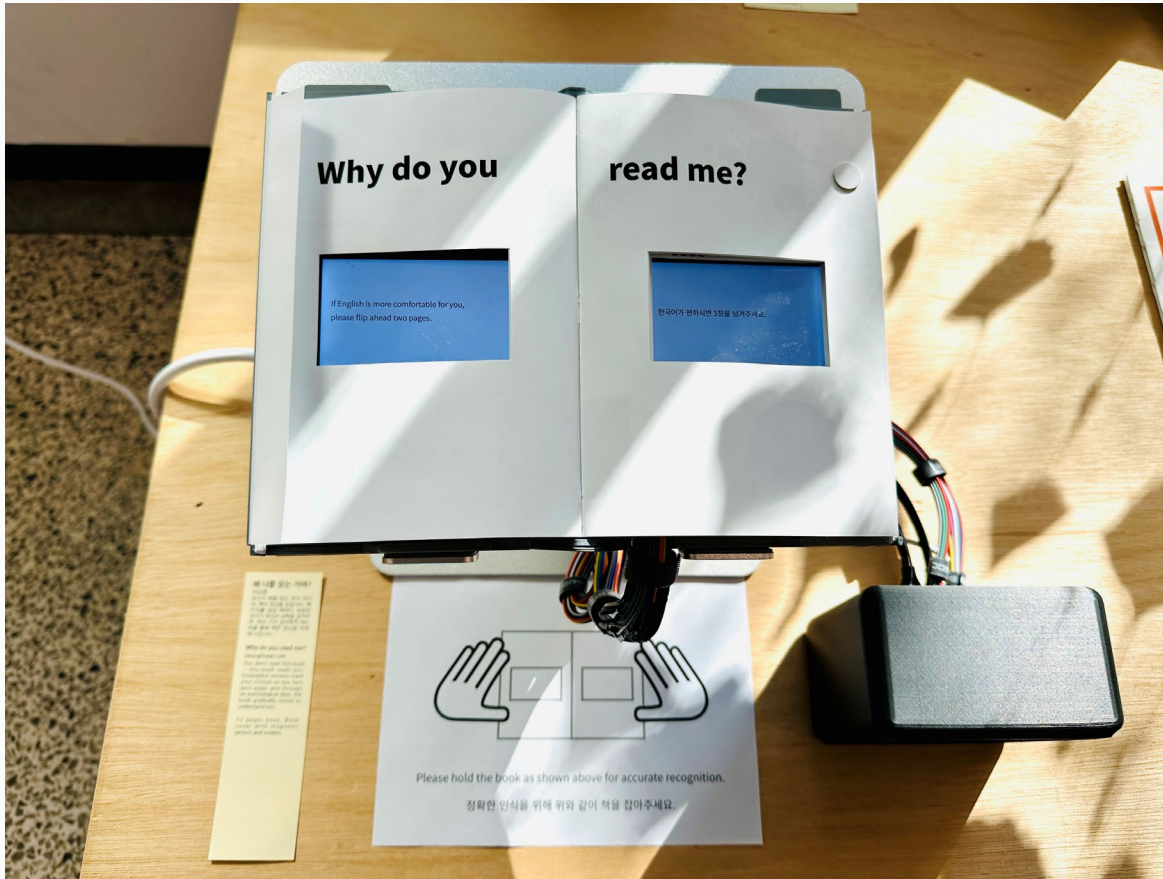
► DEMO VIDEO

<https://youtube.com/shorts/RqO9xXv41kE>

Demo video of the second prototype using a high-resolution camera and audio output.

Why do you read me?

RQ. How can physical reading behavior become input to an AI system that responds to the reader?



01 Motivation & Problem

- Traditional paper books are mostly one-directional media with fixed information, while general e-books rarely use the reader's physical reading behavior as part of the interface.
- To address this, I proposed a two-way tangible book interface that preserves the analog materiality of paper while sensing page-turning behavior and generating personalized responses.

02 System Architecture

- **Hardware Form Factor:** The system combines a sensing book-cover base with a magnet-attached paper book. The base contains Raspberry Pi, two displays, and a Hall sensor array, while the paper book remains physically replaceable.
- **Sensing & Interaction:** Page turns are detected through changes in the magnetic sensor state and converted into page position, question navigation, and selected responses.
- **LLM-based Output:** After the reading process, the accumulated choice data is interpreted by an LLM and displayed as a personalized result through the embedded screens.

Why do you read me?

RQ. How can physical reading behavior become input to an AI system that responds to the reader?



03 Design Decision

- Instead of inserting complex circuits or sensors directly into the paper book, I separated the sensing/computing book-cover base from the magnet-attached paper book.
- This structure allows different paper books to be used on the same sensing base, making the tangible form factor more expandable.

04 Research Value

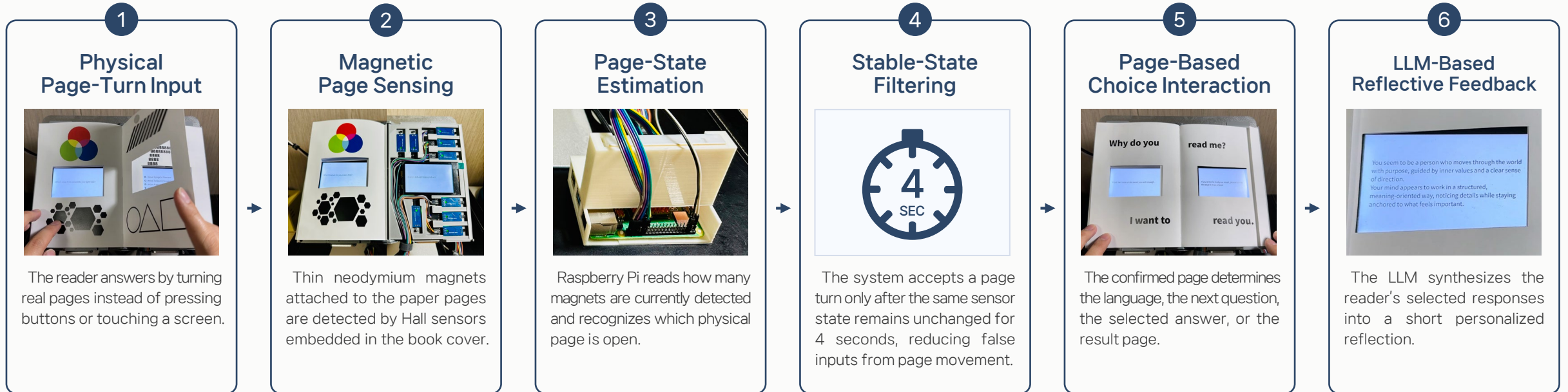
- This project explores how paper-based media can become an interactive reading interface when combined with sensing, computation, and generative AI.
- Rather than using generative AI only as a text-generation tool, the system uses it as a computational backend that interprets physical reading records and responds to them.
- By preserving familiar actions such as page turning, the system suggests an alternative to screen-centric AI interaction for users with different levels of digital literacy.
- **Next Step:** Compare tangible page-based interaction with conventional screen input across users with different levels of digital literacy.

05 Contribution

- This was an individual project. I was responsible for concept development, interaction design, hardware setup, sensing logic, LLM integration, prototyping, and documentation.
- The reflection questions and result structure were refined with feedback from a psychology major to improve their clarity and coherence.

Why do you read me?

Page-Turn Sensing & Interaction Pipeline



HCI Design Points

- Page-turning as input modality
- Open page position as computational state
- Temporal filtering for reliable interaction
- The book as an active reader of the human reader
- Reader-specific content generated from physical interaction history

Implementation

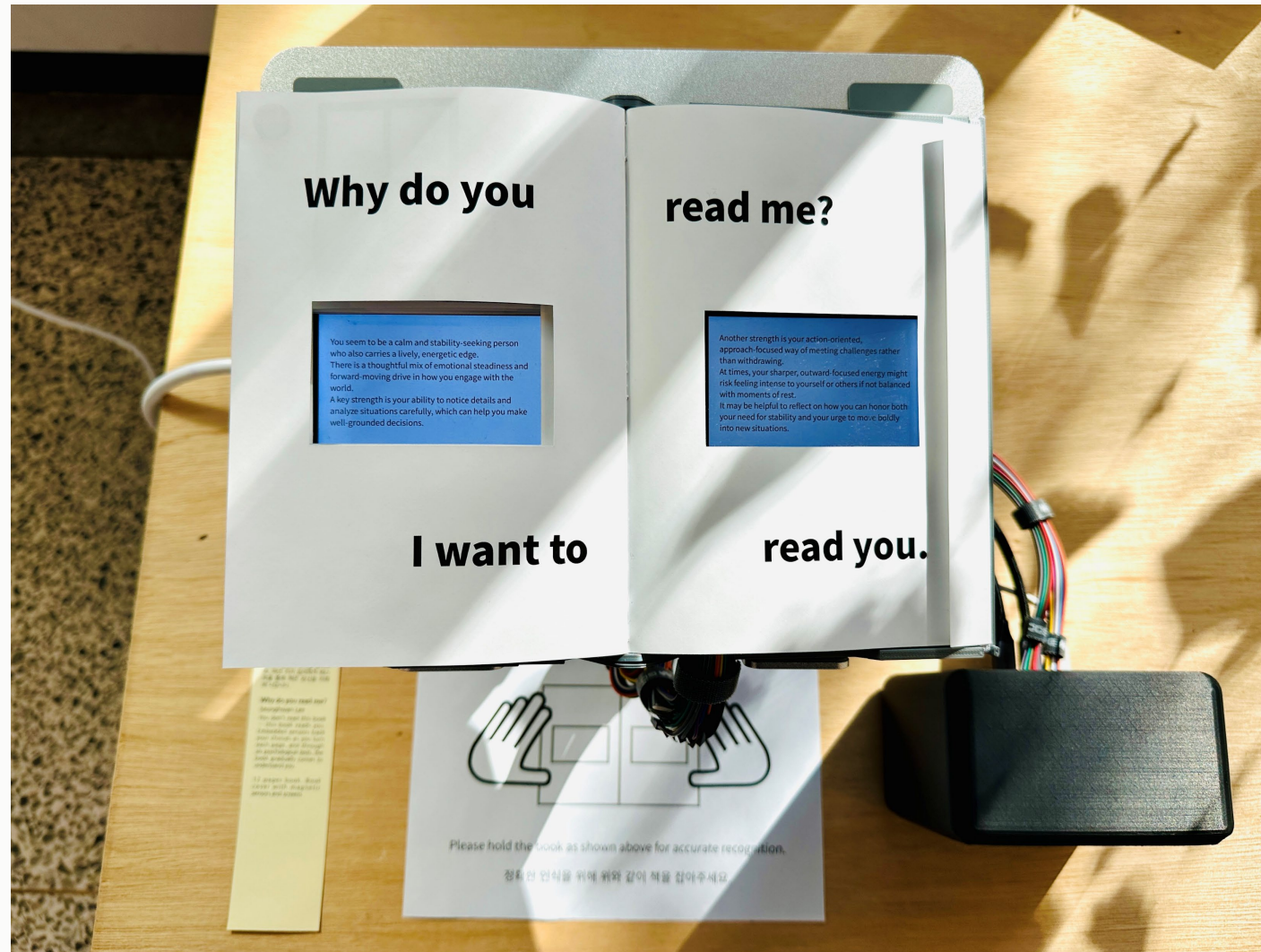
- Raspberry Pi 4B + GPIO-based sensor reading
- 9 Hall sensors + thin neodymium magnets on paper pages
- Open page estimated from the number of detected magnets
- Page-turn confirmation after the same sensor state remains stable
- Page turns used for question navigation and answer selection
- OpenAI API for final personalized reflection

Why do you read me?



The Hall sensor array in the book-cover base detects the page state of the magnet-attached paper book, converting physical page-turning into input data.

Why do you read me?



Reading behavior data collected through page turns and selected responses is transformed by the LLM into personalized reflective feedback.



Internal structure of the book-cover base integrating a Hall sensor array, dual displays, and Raspberry Pi SBC.



► DEMO VIDEO

<https://youtu.be/DdquZpPlgDg>

Page-turn interaction demo with response logging and LLM feedback.

I Need Rest

RQ. How can the act of putting a phone down become a voluntary, measurable, and playful rest interaction?



01 Motivation & Problem

- I focused on a table space originally intended for offline communication and reading, which was no longer fully serving its purpose due to smartphone use.
- Rather than simply restricting or banning smartphone use, I saw the need for an interaction that encourages users to voluntarily put down their devices and participate in offline interaction.
- To address this, I proposed a system that gives a game-like quality to the act of physically disconnecting from the smartphone, encouraging a shift in user behavior.

02 System Architecture

- **Physical Entry:** A 3D-printed smartphone bed with an NFC tag opens the rest web app when the phone is placed on it.
- **Web-based Sensing & Logging:** The web app uses the smartphone's built-in tilt sensing to measure how long the phone remains still, and saves the rest record to Firebase.
- **Public Feedback:** The saved records are visualized on a real-time leaderboard, turning individual phone-down time into shared feedback in the space.

I Need Rest

RQ. How can the act of putting a phone down become a voluntary, measurable, and playful rest interaction?



03 Design Decision

- I used the smartphone itself as the sensing device, without adding a separate motion sensor.
- This made the installation lightweight and easier to operate repeatedly in a real space.
- Rather than forcing users to stay away from their phones, the system makes putting the phone down a visible, voluntary action with immediate feedback.

04 Research Value

- This project approaches digital wellbeing not as a restriction problem, but as an interaction-design problem centered on voluntary participation and playful feedback.
- The system translates an everyday behavior—putting a smartphone down—into measurable interaction data through lightweight sensing and real-time logging.
- During **the 10-day** deployment, the system recorded **87 rest sessions** from **34 named and 9 anonymous entries**, demonstrating repeated voluntary engagement with the installation. A controlled user study would be required to determine whether such interaction leads to sustained changes in smartphone-use behavior.

05 Contribution

This was an individual project. I was responsible for problem definition, interaction design, object fabrication, NFC and web-based implementation, Firebase logging, leaderboard integration, installation, and operation.

Phone-Down Sensing & Gamified Feedback Pipeline



HCI Design Points

- Putting the phone down as a behavioral commitment
- Phone stillness as a measurable proxy for phone-down state
- Smartphone bed as a physical metaphor for phone sleep
- NFC as a low-friction bridge between installation and web interaction
- Public leaderboard as playful social motivation

Implementation

- NFC-triggered web app
- 3D-printed smartphone bed with NFC sticker
- Resting baseline captured when the timer starts
- $\pm 1^\circ$ beta/gamma movement threshold
- Firebase Realtime Database for name, activity, and duration logging
- Real-time Hall of Rest leaderboard display



Installation view showing smartphone beds and a large leaderboard, where individual rest time is shared as visual feedback in a public space.



After tapping the NFC tag and placing the phone on the bed, the web app detects the phone's stillness and measures the rest duration.



Feedback screen that saves participants' rest records to Firebase database and visualizes them on a real-time leaderboard.



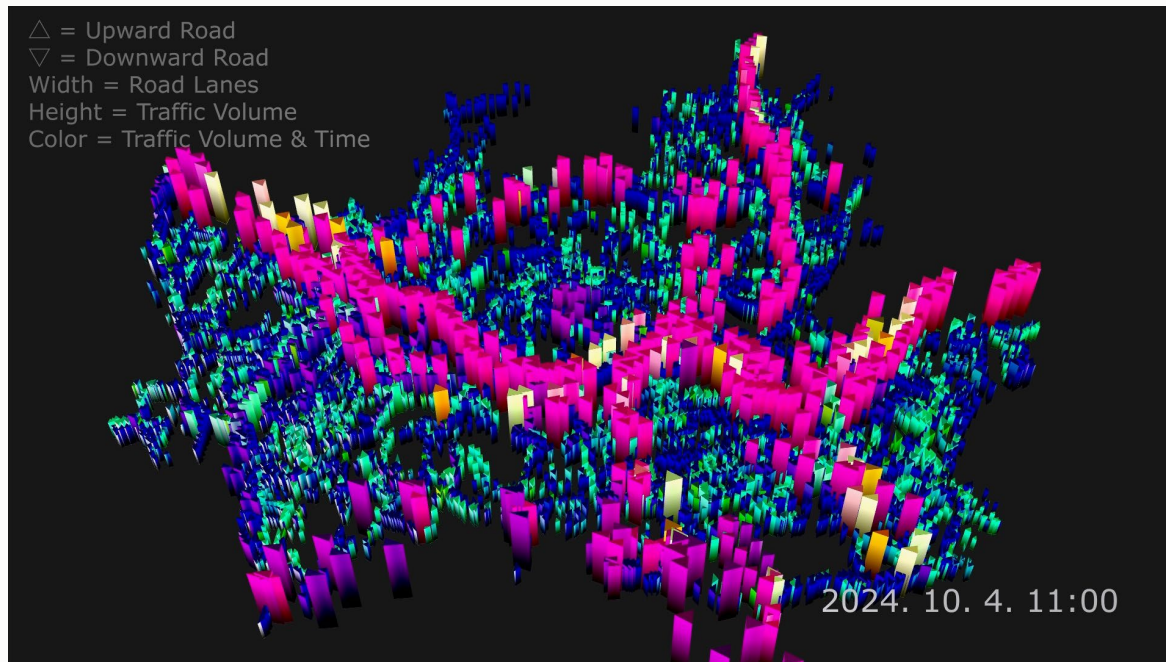
► DEMO VIDEO

<https://youtu.be/LiPgG6g4JF8>

Demo video of the phone-down interaction
: NFC tagging, rest-time measurement, movement detection, and leaderboard update.

City of Organism

RQ. How can large-scale urban time-series data be structured and transformed into a dynamic spatial representation?



01 Motivation & Concept

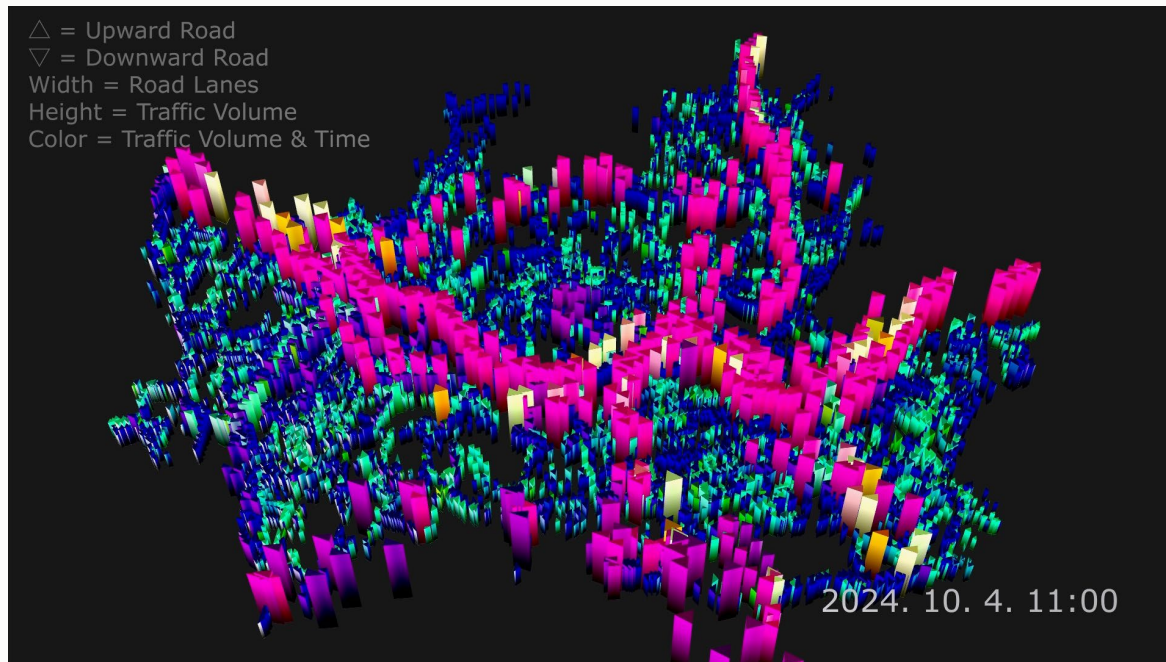
- Urban traffic flow is recorded as large-scale numerical data, but this information is usually presented through graphs or tables, making it difficult to intuitively experience.
- I focused on the structural similarity between the flow of a large city and the exchange of materials inside microscopic cells, and aimed to reinterpret traffic data through a biological metaphor.
- Based on this idea, I proposed a project that reconstructs Seoul traffic data not simply as numerical visualization, but as a 3D system with organic movement and spatial structure.

02 System Architecture

- **Data Processing:** Traffic-speed data from TOPIS and Seoul standard link mapping data were joined by road-link ID and converted into coordinate-based road segments.
- **Time-Series Structuring:** Hourly speed values were reorganized into a time-indexed TSV dataset and imported into TouchDesigner.
- **Parametric Visualization:** Lane count and traffic-speed values were mapped to prism scale, height, and color, creating an animated 3D flow of Seoul's road network.

City of Organism

RQ. How can large-scale urban time-series data be structured and transformed into a dynamic spatial representation?



03 Design Decision

- The goal was not simply to reproduce traffic data as a statistical graph, but to make the flow of the city feel like a living organism.
- To do this, I used cellular metabolism as a visual metaphor and translated traffic circulation into spatial motion, form, and color.
- Instead of mapping every value directly to movement speed, I used form and color variation to keep the visualization visually coherent.

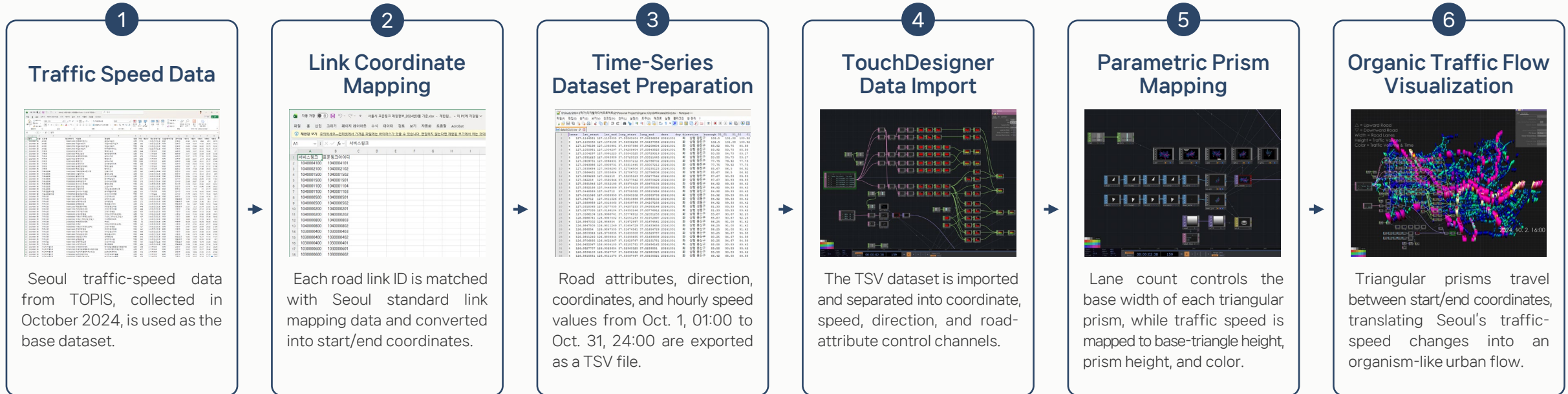
04 Research Value

- This project demonstrates the ability to work with public data and convert it into coordinate-based visual structures.
- By combining geographic data with a biological metaphor, it explores how complex urban data can be understood as spatial flow rather than static numbers.
- Through this project, I examined how data visualization can function as an interface for perceiving structure and movement.

05 Contribution

- This was an individual project. I was responsible for public data preprocessing, link ID and coordinate mapping, TouchDesigner-based 3D visualization, parameter mapping, visual metaphor design, and documentation.

Public Data Processing & Spatial Visualization Pipeline



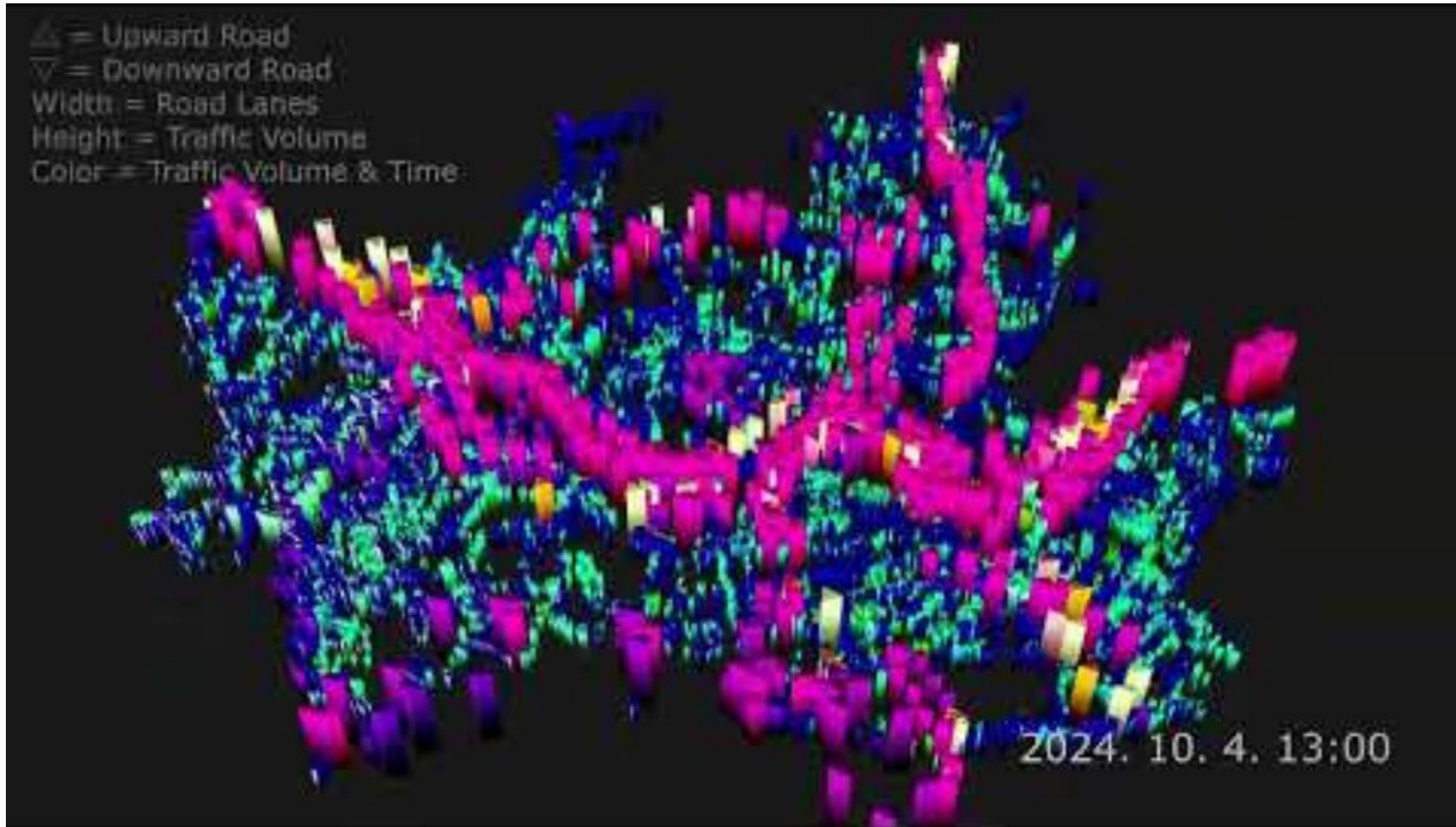
Data Visualization Points

- Time-series traffic data becomes animated spatial flow
- Road-link IDs become coordinate-based geometry
- Traffic speed becomes visual intensity and structural variation
- Lane count becomes physical scale within the visualization
- Urban flow is interpreted through a biological metaphor

Implementation

- Road-link and traffic-speed data joined by link ID
- Start/end coordinates extracted for each road segment
- Hourly speed data reorganized into a time-indexed TSV
- Traffic speed mapped to base-triangle height, prism height, and color
- Lane count mapped to prism base width
- Full October sequence played back as a loop





► DEMO VIDEO

<https://youtu.be/7Uyt15TXI0M>

Time-series visualization video showing Seoul traffic-speed data as animated 3D flow with prism height and color changes.

Part II.

Physical Interaction Concepts & Invention

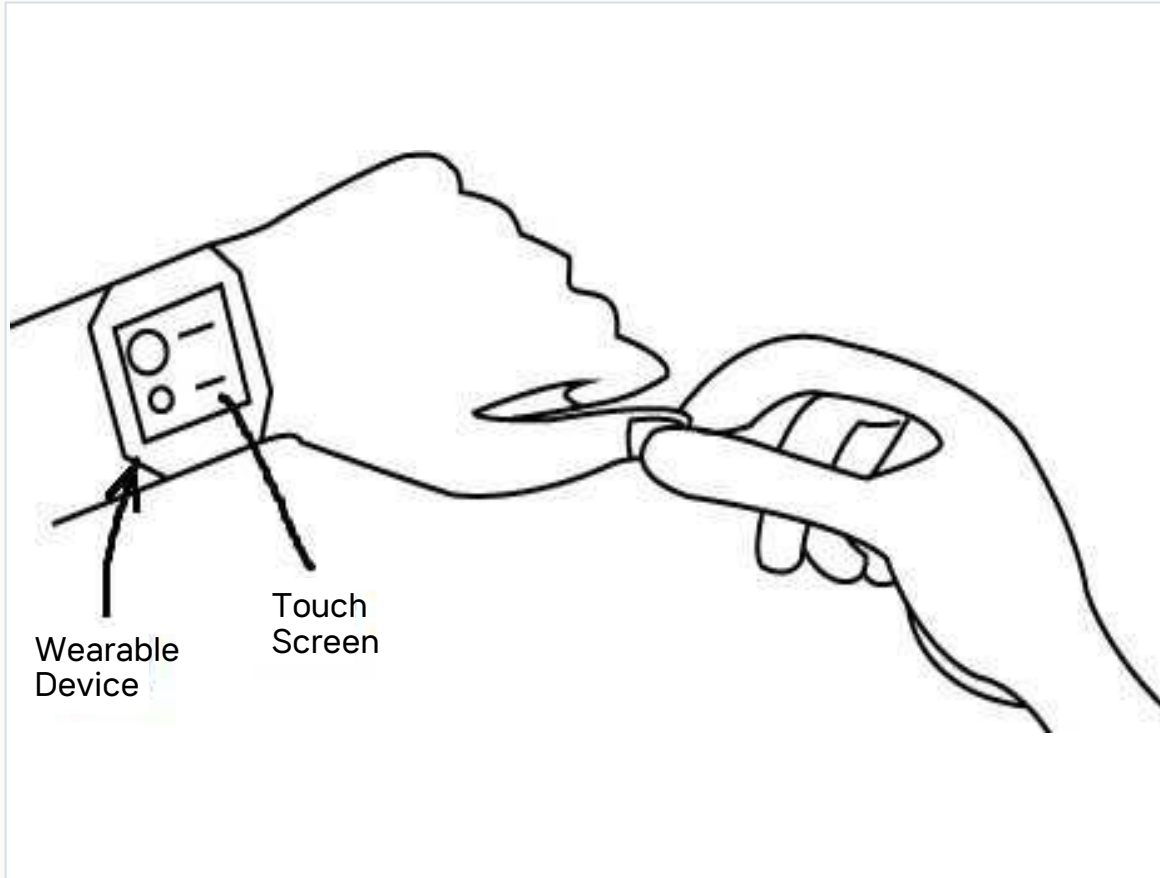
: Registered concepts exploring alternative sensing and input modalities.

Utility Model: Thumbnail Trackpad

RQ. How can small-screen wearable devices be controlled precisely by relocating the input area from the screen to an on-body surface?

Utility Model Registration No. 20-0485654

Registered as: Wearable Device Track Pad Using Infrared Light



01 Problem & Proposal

- Small wearable devices such as smartwatches have limited screen size, making precise touch control difficult. The user's finger can also occlude the screen during interaction, creating the fat finger problem.
- To provide an additional input area without increasing the device size, I proposed an on-body micro-interaction that uses the thumbnail surface as a virtual trackpad.

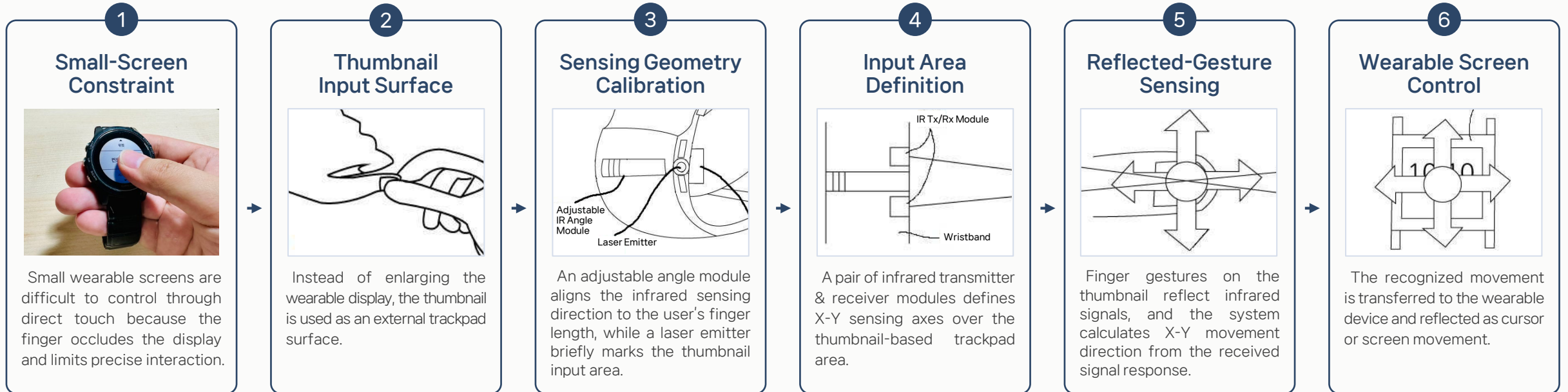
02 Core Principle

- **On-body Interface Design:** Instead of expanding the screen of the device, I designed a micro-interaction structure that uses the surface of the user's thumbnail as a virtual X-Y axis trackpad.
- **Infrared Sensing:** Infrared emitters and receivers placed on the side of the wearable device scan the position and movement of the index finger passing over the thumbnail, such as swipe and tap gestures, and convert them into input signals.

03 Research Potential

- This concept shows the potential of on-body interaction and micro-interaction design by shifting the input area from the device to the user's body without increasing the device form factor.
- In the future, it could be developed into a working hardware prototype by combining compact IR sensor arrays with machine-learning-based gesture recognition.

On-Body Infrared Trackpad Interaction Pipeline



HCI Design Points

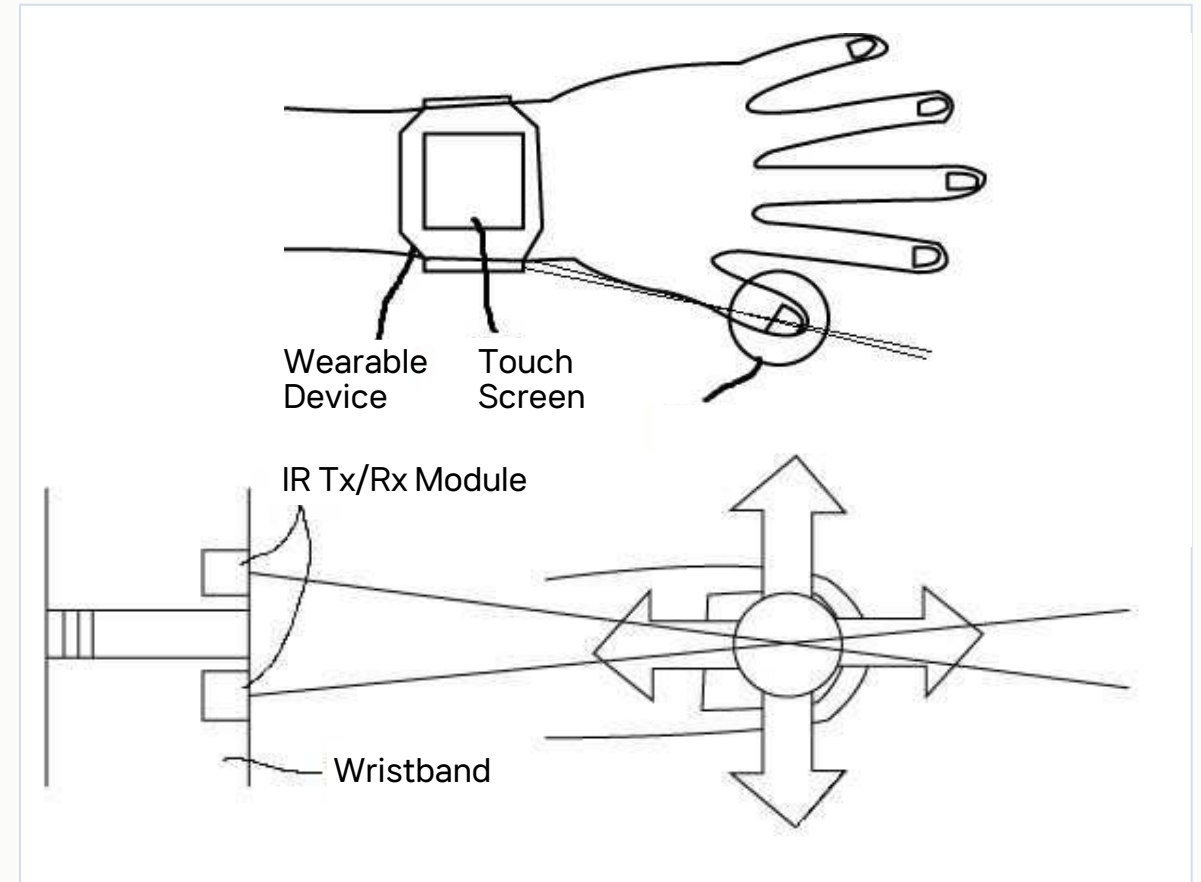
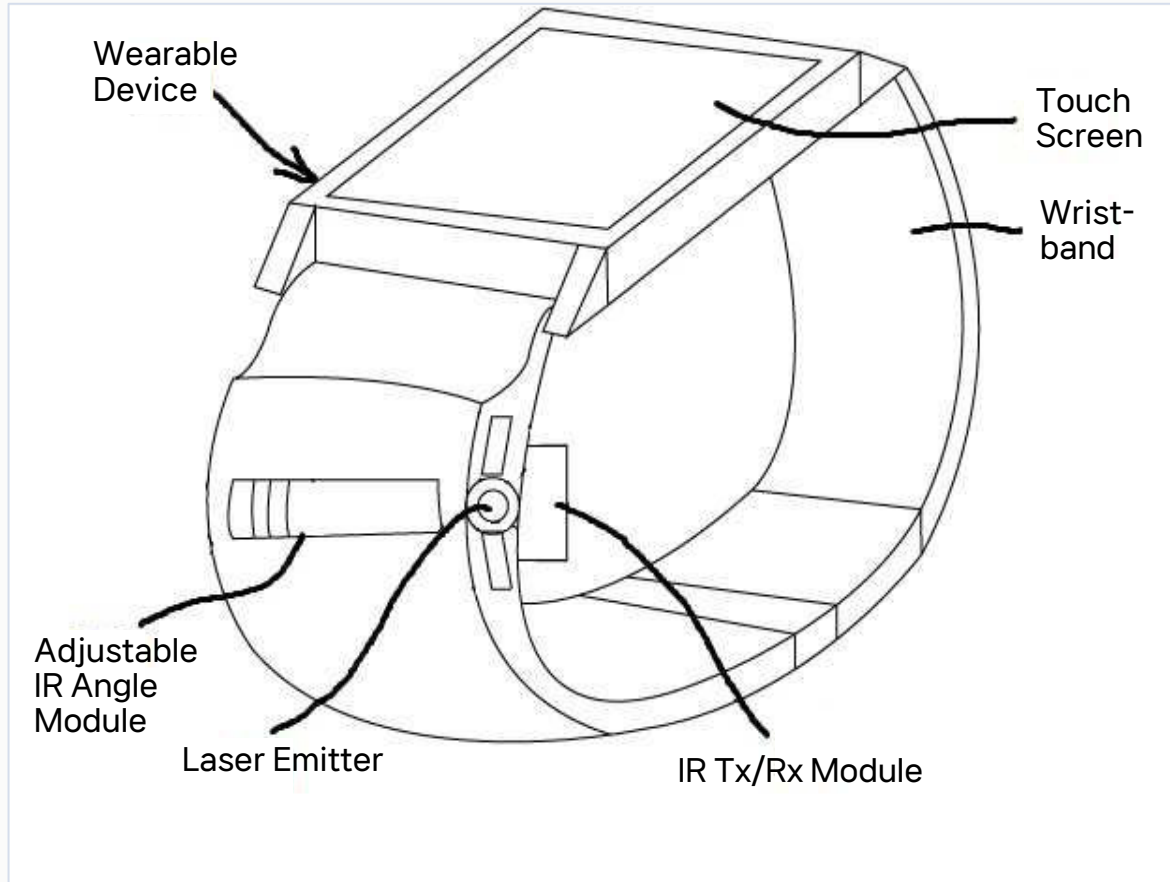
- Indirect input for small-screen wearables
- Thumbnail as an always-available on-body input surface
- Reduced screen occlusion during interaction
- Adjustable sensing geometry for different finger lengths
- Invisible infrared sensing area for gesture input

Technical Principle

- Wristband-type wearable device
- Infrared transmitter/receiver pair
- X-Y directional sensing axes
- Thumbnail-based trackpad area
- Adjustable infrared angle module
- Wireless input transmission to the wearable device

Implementation Note

As a later implementation refinement, a small auxiliary camera could be used only during initial setup to detect the thumbnail position and guide software-based infrared alignment. Real-time gesture sensing would remain infrared-based rather than vision-based.



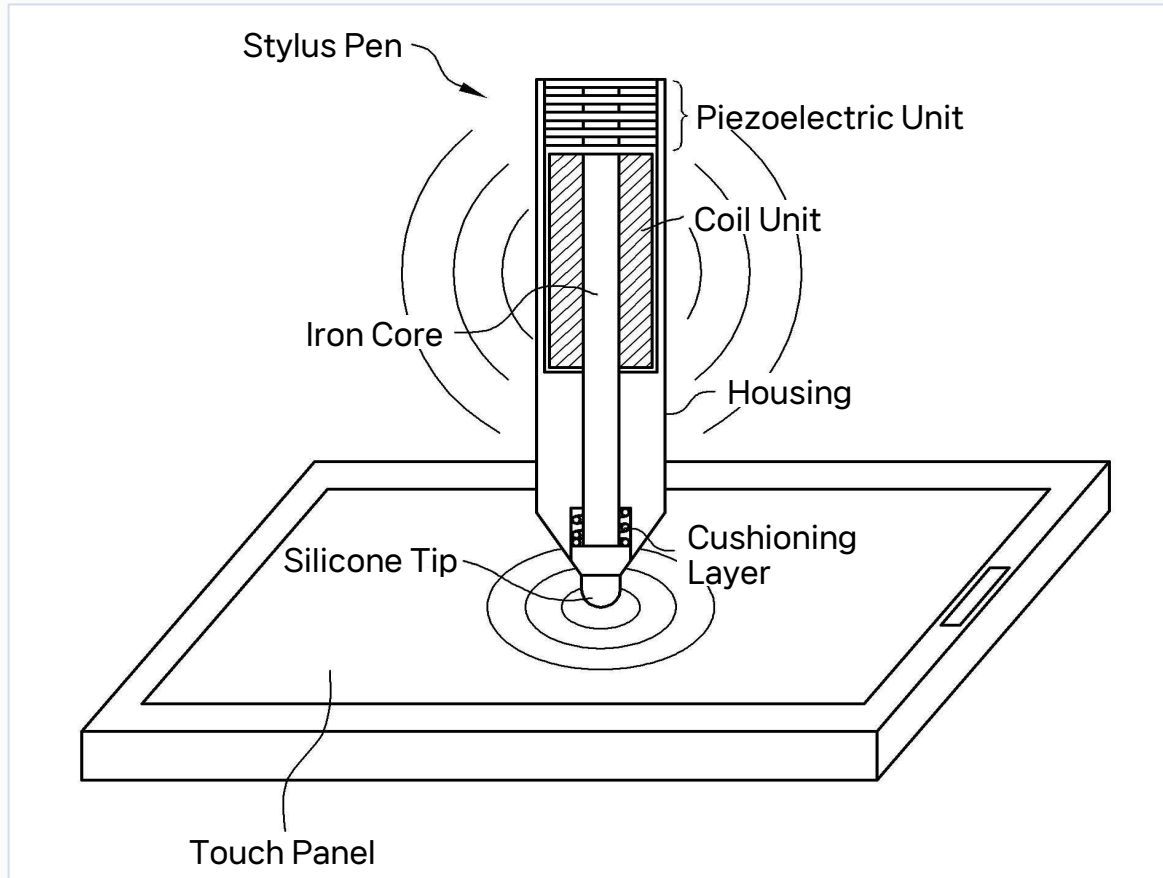
On-body Thumbnail Trackpad concept showing the wearable layout and IR-based finger-movement sensing over the thumbnail surface.

Utility Model: Battery-less Smart Stylus

RQ. How can a battery-less stylus enable pressure- and tilt-aware drawing using only writing force and a smart device's built-in sensors?

Utility Model Registration No. 20-0485653

Registered as: Stylus Pen



01 Problem & Motivation

- Active styluses that support pressure sensing usually require internal batteries and electronic circuits, which increase weight, create charging inconvenience, and raise manufacturing cost.
- To address this, I proposed a passive smart stylus that allows a smart device to recognize writing pressure and tilt without a separate power supply or complex data communication module.

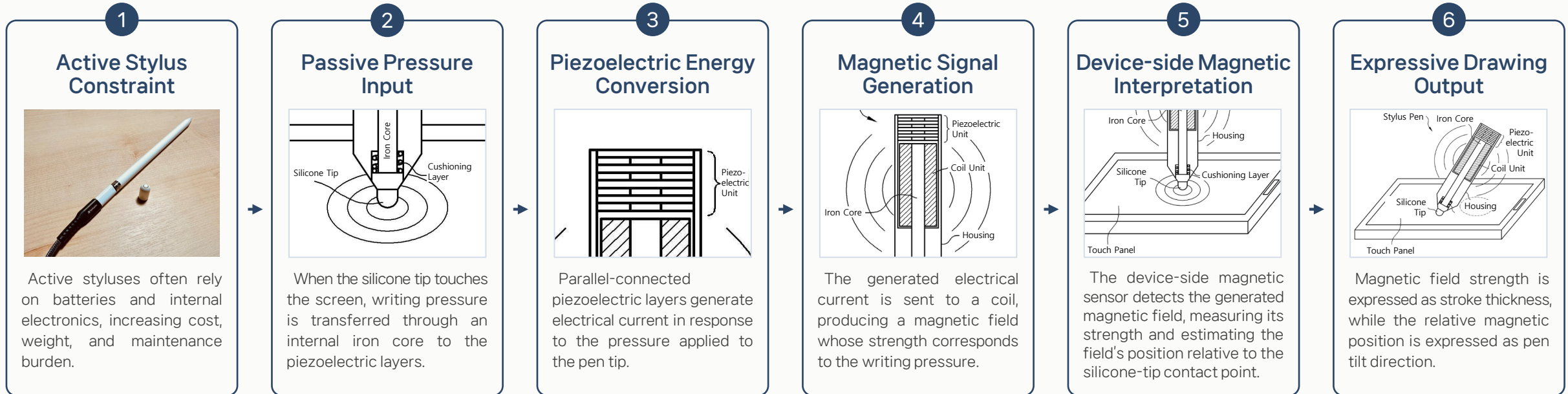
02 Core Principle

- **Piezoelectric Effect:** Piezoelectric elements are placed inside the pen to generate voltage in proportion to the physical pressure applied to the pen tip.
- **Magnetic Field Sensing:** The generated electrical signal flows through an internal coil, creating a subtle magnetic field change that corresponds to the writing pressure. A magnetic sensor built into the smart device detects this change, estimating writing pressure from field strength and pen tilt from the relative magnetic position.

03 Research Potential

- This proposal removes the need for a heavy battery and Bluetooth module, improving user experience while reducing device complexity.
- Its novelty and industrial applicability were recognized as an analog-to-digital sensing structure that converts physical force into a magnetic signal, leading to official registration as a Utility Model by the Korean Intellectual Property Office.

Battery-less Pressure-to-Magnetic Stylus Pipeline



HCI Design Points

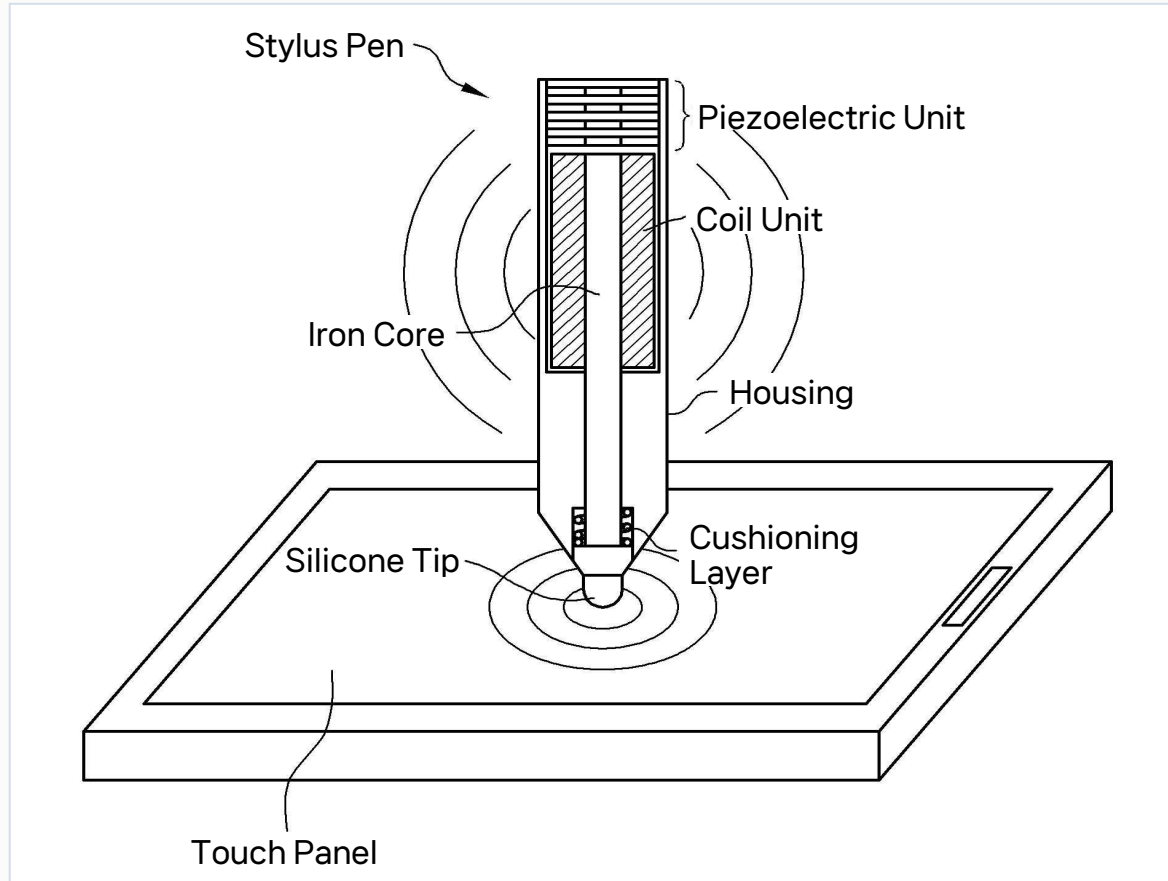
- Low-cost alternative to active stylus systems
- Battery-less stylus interaction without charging burden
- Physical writing force converted into magnetic input
- Magnetic field as a passive interaction medium
- Pressure-mapped stroke thickness for expressive drawing
- Tilt-aware drawing through relative magnetic position

Technical Principle

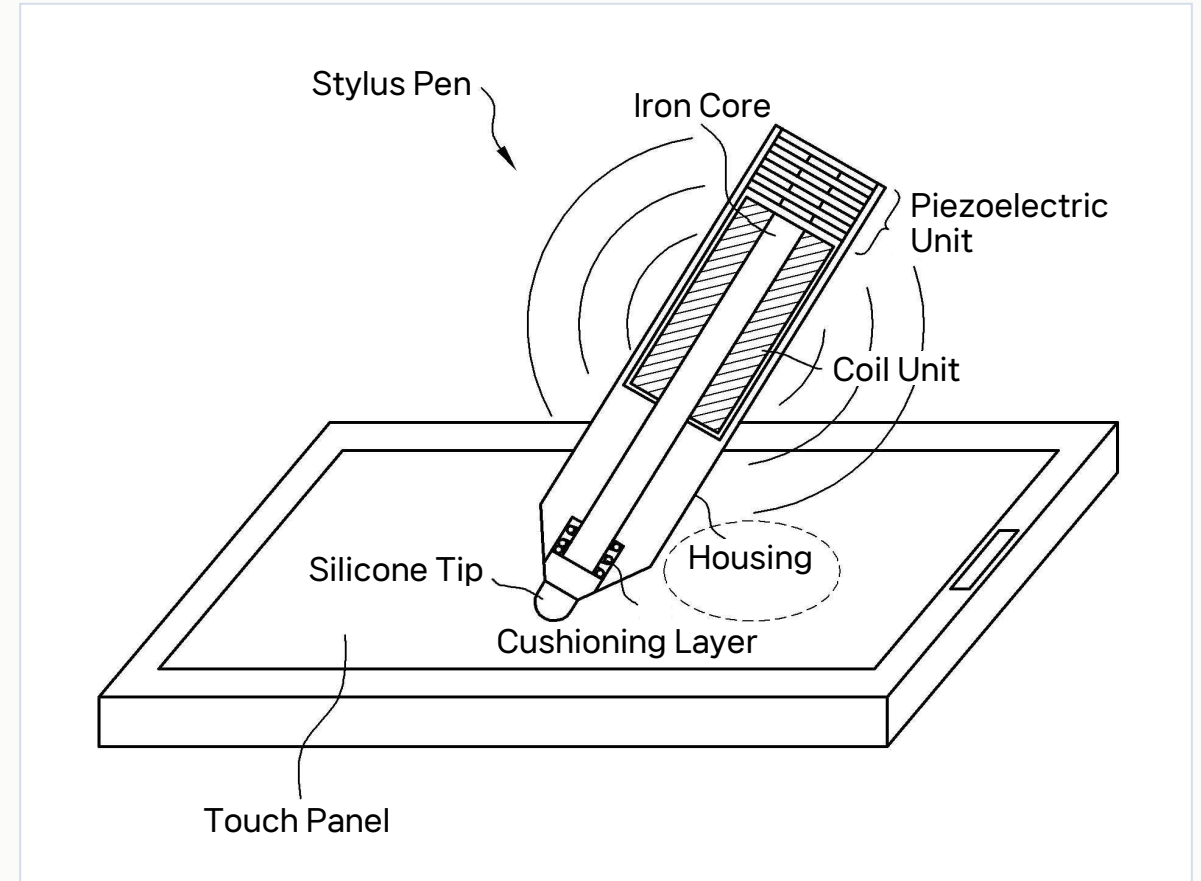
- Silicone pen tip for touch contact position
- Internal iron core for pressure transfer and magnetic amplification
- Parallel-connected layered piezoelectric elements
- Coil-based magnetic field generation
- Device-side magnetic sensing
- Magnetic field strength mapped to stroke thickness
- Relative magnetic position mapped to pen tilt

Implementation Note

A calibration step could be added before use by pressing the stylus on several predefined screen points. By recording the magnetic field pattern at each point, the system could improve magnetic source position estimation across different device sensor layouts.



Battery-less pressure sensing through piezoelectric signal generation and coil-based magnetic field changes.



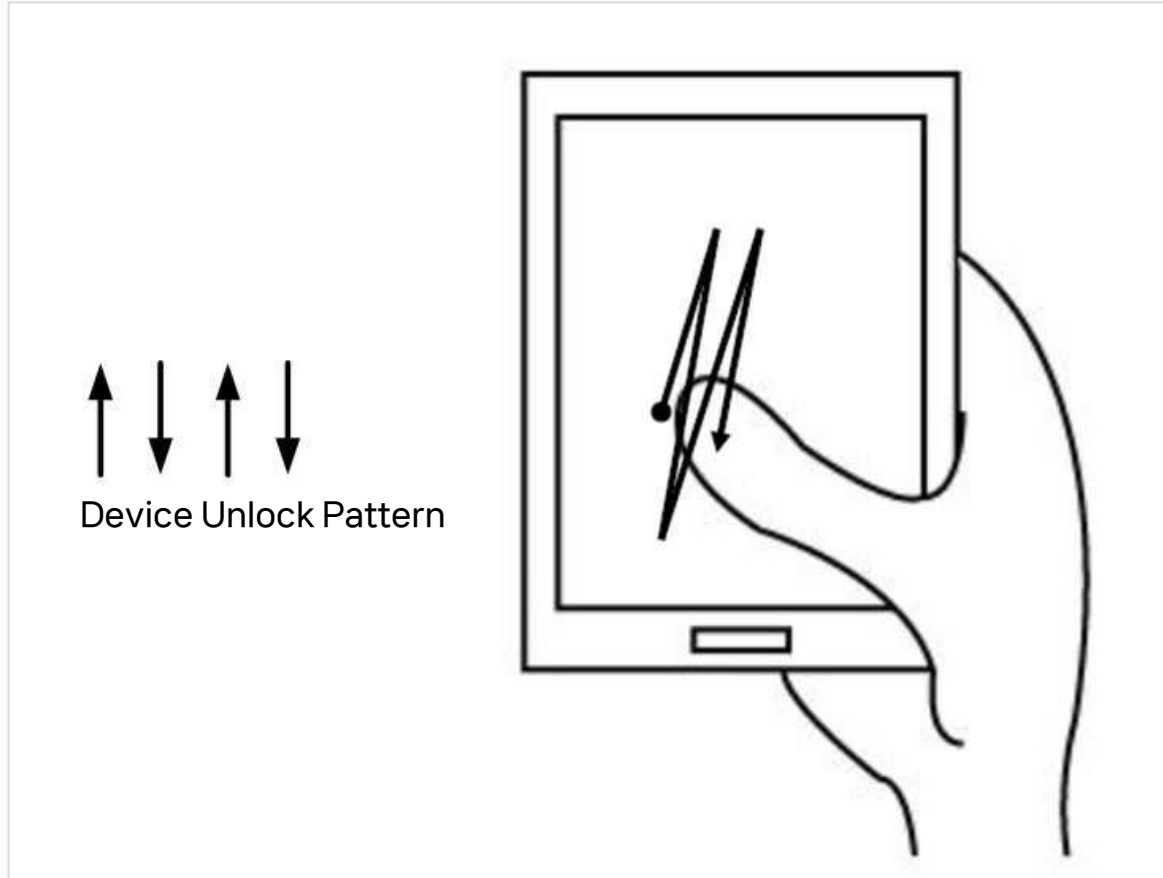
Tilt estimation through the relative position between the silicone tip contact point and the detected magnetic field.

Patent: Vector-based Gesture Unlock

RQ. How can mobile unlocking work without requiring users to connect fixed screen points?

Patent Registration No. 10-1825829

Registered as: Apparatus and Method for Unlocking Screen of Terminal



01 Problem & Proposal

- Conventional pattern-lock methods require users to connect fixed points accurately on the screen, which can become inconvenient on larger mobile displays or in one-handed use.
- They can also expose the user's input path visually or through fingerprint traces left on the screen.
- To address this, I proposed a gesture-unlock algorithm that authenticates users through direction-based gesture patterns instead of fixed point connections.

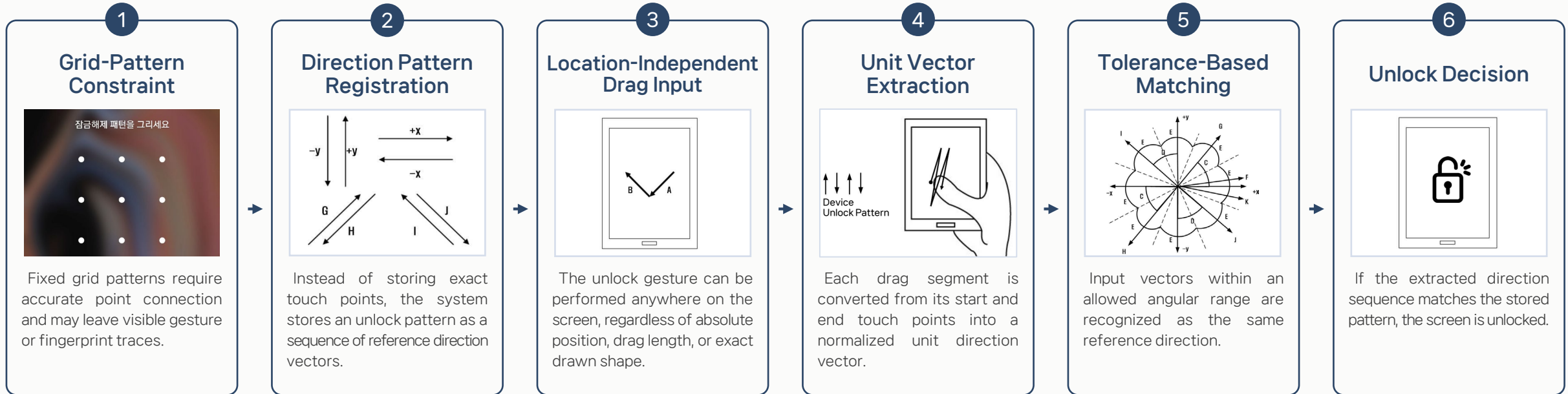
02 Core Principle

- **Vector Extraction:** When the user touches and drags on the screen, the system collects continuous X-Y touch coordinates and converts each drag segment into a direction vector connecting the start and end points.
- **Pattern Matching:** Instead of relying on fixed screen positions, the system compares the relative angles and order of multiple gesture vectors with a pre-stored unlock pattern to perform authentication.

03 Research Potential

- This concept offers a high degree of freedom because the gesture can be entered anywhere on the screen. It also has potential applications for visually impaired users or one-handed use situations.
- In the future, it could be expanded by evaluating gesture recognition accuracy, error rates, and user memory burden in an actual mobile interface environment.

Direction-Vector Unlock Interaction Pipeline



HCI Design Points

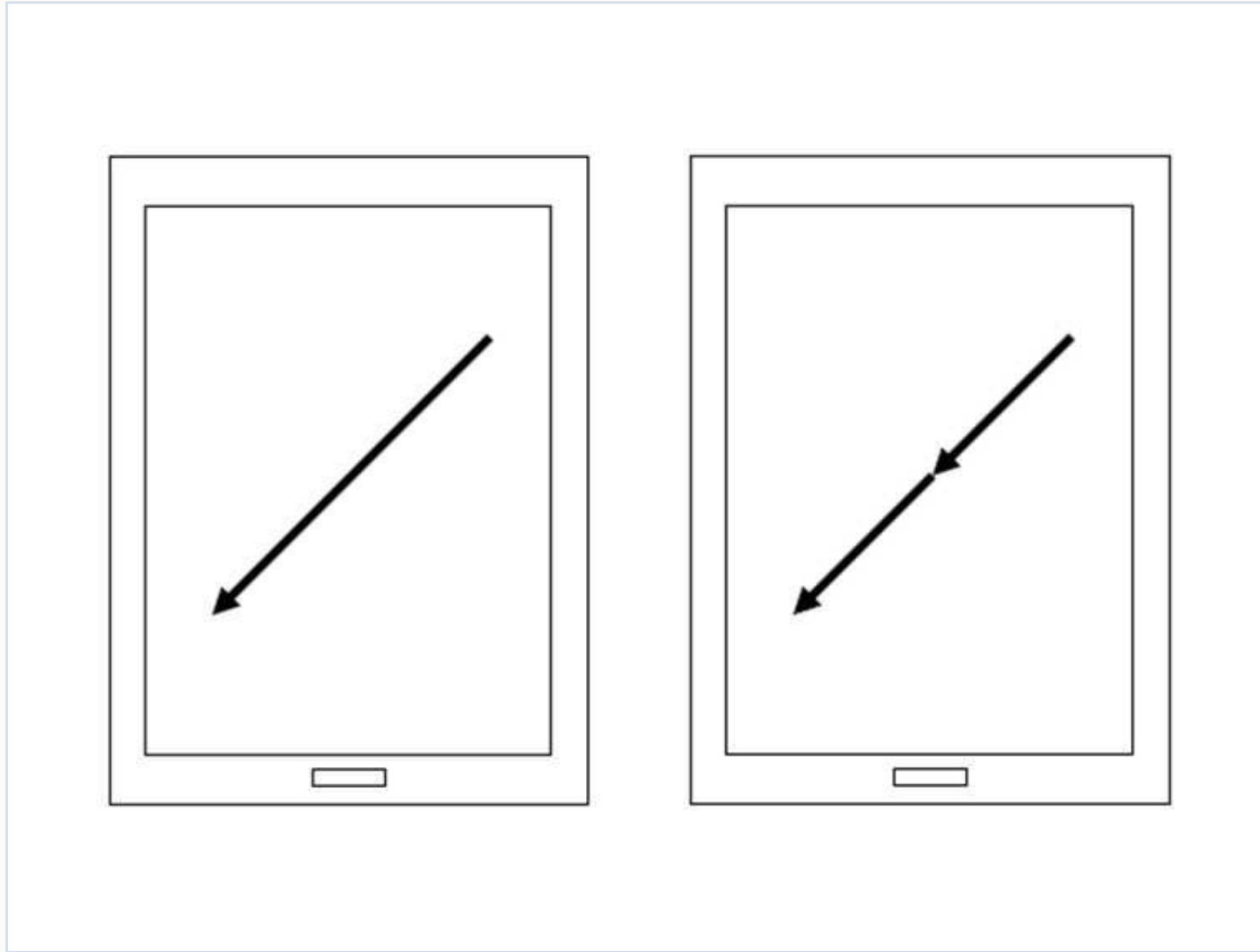
- Direction-based authentication instead of fixed-point input
- Location-independent gesture interaction
- Tolerance-based recognition for input flexibility
- Unlocking without precise point targeting
- Reduced dependence on visible or repeated gesture paths

Technical Principle

- Stored unlock pattern as a unit-vector sequence
- Reference directions on the screen x-y plane
- Start/end touch points converted into normalized vectors
- Allowed-angle matching range for each reference direction
- Full-screen, location-independent drag area
- Consecutive same-direction drags treated as a single input

Implementation Note

The angular tolerance can be adjusted to balance usability and security: a wider range makes input easier, while a narrower range increases pattern specificity.



Same-direction drag segments are merged into one input so that a continuous gesture is not counted multiple times when briefly interrupted.

Epilogue.

Research Motivation & Human Context

: From lived experience to human-centered AI and interactive systems

Research Motivation & Human Context

From lived experience to human-centered AI and interactive systems



Color Purple documentary project

01 Background

- I have been interested in people who live within our society but remain outside easy visibility.
- In particular, I focused on the lives of North Korean defectors, their sense of hometown, and the memories and emotions that remain outside public narratives.
- This interest shaped my concern with understanding human experience, emotion, memory, and sensation before approaching technology.

02 Action & Achievement

- In the documentary project *Color Purple*, I recorded the life paths and memories of North Korean defectors, focusing on individual narratives and emotions beyond ideological categories.
- This work received the Best Print award at the department's 2024 Open Studio.
- I later continued this concern through a portrait project for elderly North Korean defectors living alone, which I co-planned and photographed in collaboration with the Gyeonggi Northwest Hana Center. The project was later covered by Goyang Newspaper.

Research Motivation & Human Context

From lived experience to human-centered AI and interactive systems

고양신문

뉴스 63 지방선거 오피니언 동영상 구독신청 후원하기 여론광장

뉴스 > 사회·교통·환경

꽃그네 위에서 한 생애가 환하게 피어났다

권구영 기자 | 입력 2026.05.22 06:31 | 댓글 0

경기서북부하나센터, 북한이탈주민 장수 사진 촬영 행사 열어



2026년 지역통합지원사업 장수사진 촬영-꽃처럼 빛나는 오늘' 참가자 기념 촬영 [사진제공 = 경기서북부하나센터]

[고양신문] 지난 8일 오후, 고양시 덕양구 선유동에 자리한 치유농장 온유원의 꽃그네는 다양한 한복 색으로 물들었다. 화관을 쓴 어르신들이 꽃다발을 손에 들고 카메라를 향해 미소를 지을 때마다 한 생애가 천천히 화면 위로 떠올랐다. 카메라 셔터는 빛을 가두는 장치 가 아니라 한 사람의 시간이 여전히 여기 있음을 알리고 또 그 시간이 존중받아 마땅함을 증언하는 의식이었다.

촬영을 맡은 중앙대학교 예술대학 사진 전공 이승환 자원봉사자는 "렌즈를 통해 어르신들과 눈을 맞출 때마다 마음이 따뜻해졌다"며 "단순히 멋진 사진을 찍는 것이 아니라 한 분 한 분의 표정과 그 속에 담긴 삶의 흔적을 담아내고 싶었다"고 말했다. 그가 매번 셔터를 누르기 전 잠시 호흡을 고르던 모습은 이 작업이 기술의 영역이 아니라 예의의 영역에 가깝다는 것을 보여주었다.



중앙대학교 예술대학 사진 전공 이승환 자원봉사자가 사진을 찍고 있다. [사진제공 = 경기서북부하나센터]

실내 촬영에 이어 마련된 실외 촬영에서는 온유원 꽃그네가 배경이 됐다. 어르신들은 화관을 쓰고 꽃다발을 든 채 환하게 웃었고 차를 기다리는 동안에는 서로의 사진을 찍어 주며 시간을 보냈다.

어버이날 카네이션 바꾸기는 더한걸음행복플러스 이은주 대표가 손수 준비해 전달했다. 가족이 멀리 있거나 곁에 없는 이들에게 건네진 카네이션은 단지 꽃이 아니라 당신이 걸 어온 시간은 존중받아 마땅하다는 지역사회의 응답이었다.

이날 행사는 온유원의 장소 지원, 대학생의 재능기부, 자원봉사자들의 손길, 한복 기부, 카네이션 후원 등 여러 갈래의 마음이 모여 완성됐다. 김영숙 센터장을 비롯해 어르신 9명, 온유원 김민주 대표, 더한걸음행복플러스 이은주 대표, 중앙대 이승환 자원봉사자 등 23명이 참여했다.

03 Research Connection

- These documentary and community-based experiences strengthened my view that technology should be grounded in people's lived experiences, needs, and barriers. This perspective now informs my interest in human-centered AI and interactive systems that respond to people's needs and barriers by translating human behavior, sensory information, and context into meaningful interaction.

04 Future Research Direction

- First, I want to explore human-centered AI systems that interpret everyday behavior and context, particularly for accessibility and wellbeing.
- Second, I want to investigate multimodal interactions that translate information across visual, auditory, tactile, and physical forms.
- Third, building on my experience in sensing and physical prototyping, I want to connect AI with real-world human action beyond conventional screen- and chat-based interfaces.
- Ultimately, I aim to design and rigorously evaluate interactive systems grounded in how people perceive, behave, and live.

Press coverage of a portrait project co-planned with the Gyeonggi Northwest Hana Center.

Source: Goyang Newspaper

<https://www.mygoyang.com/news/articleView.html?idxno=88789>